

CohortConstructor

Build and Curate Study Cohorts in R Using the OMOP Common Data Model

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Introduction

- CohortConstructor package is designed to support cohort building pipelines in R, using data mapped to the OMOP Common Data Model.
- The code is publicly available in OHDSI's GitHub repository **CohortConstructor**.
- CohortConstructor v0.6.3 is available in **CRAN**.
- Vignettes with further information can be found in the package **website**.

OMOP Cohorts in R

OMOP Cohorts in R

- The `<cohort_table>` class is defined in the R package **omopgenerics**.
- This is the class that **CohortConstructor** uses, as well as other OMOP analytical packages.
- As defined in **omopgenerics**, a `<cohort_table>` must have at least the following 4 columns (without any missing values in them):
 - **cohort_definition_id**: Unique identifier for each cohort in the table.
 - **subject_id**: Unique patient identifier.
 - **cohort_start_date**: Date when the person enters the cohort.
 - **cohort_end_date**: Date when the person exits the cohort.

OMOP Cohorts in R

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```
cdm$cohort
```

```
# A query:  ?? x 4
# Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R 4.6.1//tmp/RtmptNjdPT/file21d12c3e4dc5.duckdb]
  cohort_definition_id subject_id cohort_start_date cohort_end_date
                <int>      <int> <date>              <date>
1                   1         153 1945-08-10          1945-08-19
2                   1         299 1965-10-29          1965-11-11
3                   1         439 1979-07-24          1979-08-02
4                   1         653 1959-02-01          1959-02-11
5                   1         752 1985-11-21          1985-12-04
6                   1         883 2005-04-26          2005-05-07
7                   1        1068 2018-10-21          2018-11-02
8                   1        1623 2002-05-12          2002-05-22
9                   1       2450 1980-01-17          1980-01-28
10                  1       3399 1996-11-02          1996-11-10
... .
```

OMOP Cohorts in R

Additionally, the `<cohort_table>` object has the following **attributes**:

- **Settings**: Relate each cohort definition ID with a cohort name and other variables that define the cohort.

```
settings(cdm$cohort)
```

```
# A tibble: 2 × 4
  cohort_definition_id cohort_name      cdm_version vocabulary_version
      <int> <chr>                <chr>          <chr>
1           1 viral_pharyngitis 5.3          v5.0 18-JAN-19
2           2 viral_sinusitis  5.3          v5.0 18-JAN-19
```


OMOP Cohorts in R

- **Attrition:** Store information on each inclusion criteria applied and how many records and subjects were kept after.

```
attrition(cdm$cohort)
```

```
# A tibble: 8 × 7
```

	cohort_definition_id	number_records	number_subjects	reason_id	reason	excluded_records
excluded_subjects						
	<int>	<int>	<int>	<int>	<chr>	<int>
<int>						
1	1	10217	2606	1	Initial qualifying events	0
0						
2	1	10217	2606	2	Record in observation	0
0						
3	1	10217	2606	3	Not missing record date	0
0						
4	1	10217	2606	4	Merge overlapping records	0
0						
5	2	17268	2686	1	Initial qualifying events	0
-						

OMOP Cohorts in R

- **Cohort count:** Number of records and subjects for each cohort.

```
cohortCount(cdm$cohort)
```

```
# A tibble: 2 × 3
```

	cohort_definition_id	number_records	number_subjects
	<int>	<int>	<int>
1	1	10217	2606
2	2	17268	2686

OMOP Cohorts in R

- **Cohort codelist:** Codelists used to define entry events and inclusion criteria for each cohort.

```
attr(cdm$cohort, "cohort_codelist")
```

```
# A query: ?? x 4
# Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R 4.6.1//tmp/RtmpNjdPT/file21d12c3e4dc5.duckdb]
  cohort_definition_id codelist_name      concept_id codelist_type
                <int> <chr>              <int> <chr>
1                   1 viral_pharyngitis   4112343 index event
2                   2 viral_sinusitis     40481087 index event
```

CohortConstructor function Sets

Base cohorts *Cohort*

construction based on clinical concepts or demographics.

Requirements and Filtering

Demographic restrictions, event presence/absence conditions, and filtering specific records.

Update cohort entry and exit

Adjusting entry and exit dates to align with study periods, observation windows, or key events.

Transformation and

Combination *Merging, stratifying, collapsing, matching, or intersecting cohorts.*

Base cohorts

Functions to build base cohorts

- `demographicsCohort()`
- `conceptCohort()`
- `measurementCohort()`
- `deathCohort()`

Create the cdm_reference object

```
# Load relevant packages
library(CDMConnector)
library(CodeListGenerator)
library(CohortConstructor)
library(CohortCharacteristics)
library(duckdb)
library(DBI)
library(dplyr)
library(gt)
library(here)

con <- dbConnect(drv = duckdb(dbdir = here("Databases", "GiBleed.duckdb")))

cdm <- cdmFromCon(
  con = con,
  cdmSchema = "main",
  writeSchema = "results",
  # ... other arguments ...
)
```

Demographics based - Example

- Two cohorts, females and males, both aged 18 to 60 years old, with at least 365 days of previous observation in the database.

```
cdm$age_cohort <- demographicsCohort(  
  cdm = cdm,  
  ageRange = c(18, 60),  
  sex = c("Female", "Male"),  
  minPriorObservation = 365,  
  name = "age_cohort"  
)  
  
settings(x = cdm$age_cohort)
```

A tibble: 2 × 5

	cohort_definition_id	cohort_name	age_range	sex	min_prior_observation
	<int>	<chr>	<chr>	<chr>	<dbl>
1	1	demographics_1	18_60	Female	365
2	2	demographics_2	18_60	Male	365

Demographics based - Example

```
cohortCount(cohort = cdm$age_cohort)
```

```
# A tibble: 2 × 3
  cohort_definition_id number_records number_subjects
      <int>           <int>           <int>
1             1         1373         1373
2             2         1321         1321
```

```
attrition(x = cdm$age_cohort)
```

```
# A tibble: 12 × 7
  cohort_definition_id number_records number_subjects reason_id reason
excluded_records excluded_subjects
      <int>           <int>           <int>      <int> <chr>
<int>      <int>
1             1         2694         2694        1 Initial qualifying events
0             0
2             1         2694         2694        2 Non-missing sex
0             0
3             1         1373         1373        3 Sex requirement: Female
1321        1321
4             1         1373         1373        4 Non-missing year of birth
0             0
5             1         1373         1373        5 Age requirement: 18 to 60
```

Demographics based - Example

To better visualise the **attrition**, we can use the package **CohortCharacteristics** to create a **formatted table**:

```
cdm$age_cohort |>
  summariseCohortAttrition() |>
  tableCohortAttrition()
```

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
Synthea; demographics_1				
Initial qualifying events	2,694	2,694	0	0
Non-missing sex	2,694	2,694	0	0
Sex requirement: Female	1,373	1,373	1,321	1,321
Non-missing year of birth	1,373	1,373	0	0
Age requirement: 18 to 60	1,373	1,373	0	0
Prior observation requirement: 365 days	1,373	1,373	0	0
Synthea; demographics_2				
Initial qualifying events	2,694	2,694	0	0
Non-missing sex	2,694	2,694	0	0
Sex requirement: Male	1,321	1,321	1,373	1,373
Non-missing year of birth	1,321	1,321	0	0
Age requirement: 18 to 60	1,321	1,321	0	0
Prior observation requirement: 365 days	1,321	1,321	0	0

Concept based - Example

Let's create a cohort of medications that contains two drugs: diclofenac, and acetaminophen.

1. Create codelist of the drug of interest

```
drugCodes <- list(  
  "diclofenac" = c(1124300L),  
  "acetaminophen" = c(1125315L, 1127078L, 1127433L, 19133768L, 40162522L, 40229134L, 40231925L)  
) |>  
  newCodelist()
```

Concept based - Example

2. Create concept based cohorts

```
cdm$medications <- conceptCohort(  
  cdm = cdm,  
  conceptSet = drugCodes,  
  name = "medications"  
)
```

```
settings(x = cdm$medications)
```

```
# A tibble: 2 × 4
```

	cohort_definition_id	cohort_name	cdm_version	vocabulary_version
	<int>	<chr>	<chr>	<chr>
1	1	acetaminophen	5.3	v5.0 18-JAN-19
2	2	diclofenac	5.3	v5.0 18-JAN-19

Concept based - Example

- Attrition

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
acetaminophen				
Initial qualifying events	14,205	2,679	0	0
Record in observation	14,205	2,679	0	0
Not missing record date	14,205	2,679	0	0
Merge overlapping records	13,908	2,679	297	0
diclofenac				
Initial qualifying events	850	850	0	0
Record in observation	830	830	20	20
Not missing record date	830	830	0	0
Merge overlapping records	830	830	0	0

Concept based - Example

- Cohort codelist as an attribute

```
attr(cdm$medications, "cohort_codelist")
```

```
# A query: ?? x 4
# Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R
4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]
  cohort_definition_id codelist_name concept_id codelist_type
      <int> <chr>          <int> <chr>
1             1 acetaminophen   1125315 index event
2             1 acetaminophen   1127078 index event
3             1 acetaminophen   1127433 index event
4             1 acetaminophen  19133768 index event
5             1 acetaminophen   40162522 index event
6             1 acetaminophen   40229134 index event
7             1 acetaminophen   40231925 index event
8             2 diclofenac       1124300 index event
```



Your turn!

Get Started: Create the cdm_reference object

```
# Load relevant packages
library(CDMConnector)
library(CodeListGenerator)
library(CohortConstructor)
library(CohortCharacteristics)
library(duckdb)
library(DBI)
library(dplyr)
library(gt)
library(here)

con <- dbConnect(drv = duckdb(dbdir = here("Databases", "delphi.duckdb")))
cdm <- cdmFromCon(
  con = con,
  cdmSchema = "main",
  writeSchema = "results",
  writePrefix = "cc_"
)
```




Exercise 1 - Base cohorts

Create a base cohort table named **base_conditions_cohort** that has two cohorts: **Rheumatoid Arthritis (RA)** and **Pneumonia**. Use the codelist below.

```
codes <- list(  
  "ra" = c(80809L),  
  "pneumonia" = c(255848L)  
) |> newCodelist()
```

►  Click to see solution

Move to the next slide to see the cohort attrition



Exercise 1 - Base cohorts

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
Delphi-2M; pneumonia				
Initial qualifying events	783	783	0	0
Record in observation	783	783	0	0
Not missing record date	783	783	0	0
Merge overlapping records	783	783	0	0
Delphi-2M; ra				
Initial qualifying events	676	676	0	0
Record in observation	676	676	0	0
Not missing record date	676	676	0	0
Merge overlapping records	676	676	0	0

Requirements and Filtering

Functions to apply requirements and filter

- On demographics
 - `requireDemographics()`
 - `requireAge()`
 - `requireSex()`
 - `requirePriorObservation()`
 - `requireFutureObservation()`
- On cohort entries
 - `requireIsFirstEntry()`
 - `requireIsLastEntry()`
 - `requireIsEntry()`
- Require presence or absence based on other cohorts, concepts, and tables
 - `requireCohortIntersect()`
 - `requireConceptIntersect()`
 - `requireTableIntersect()`
- Other
 - `requireInDateRange()`
 - `requireMincohortCount()`
 - `requireDuration()`

Requirement functions - Example

- We can apply different inclusion criteria using CohortConstructor's functions in a pipe-line fashion. For instance, in what follows we require
 - only first record per person
 - subjects 18 years old or more at cohort start date
 - only females
 - at least 30 days of prior observation at cohort start date

```
cdm$medications_requirement <- cdm$medications |>
  requireIsFirstEntry(name = "medications_requirement") |>
  requireDemographics(
    ageRange = list(c(18, 150)),
    sex = "Female",
    minPriorObservation = 30,
    name = "medications_requirement"
  )
```

Requirement functions - Example

Attrition Acetaminophen

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
acetaminophen				
Initial qualifying events	14,205	2,679	0	0
Record in observation	14,205	2,679	0	0
Not missing record date	14,205	2,679	0	0
Merge overlapping records	13,908	2,679	297	0
Restricted to first entry	2,679	2,679	11,229	0
Age requirement: 18 to 150	309	309	2,370	2,370
Sex requirement: Female	176	176	133	133
Prior observation requirement: 30 days	176	176	0	0
Future observation requirement: 0 days	176	176	0	0

Requirement functions - Example

- Now, we only want to keep those records for which prescription happened on the same day of a healthcare visit.

```
cdm$medications_requirement <- cdm$medications_requirement |>
  requireTableIntersect(
    tableName = "visit_occurrence",
    indexDate = "cohort_start_date",
    window = c(0, 0),
    intersections = c(1, Inf)
  )
```

Requirement functions - Example

Attrition Acetaminophen

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
acetaminophen				
Initial qualifying events	14,205	2,679	0	0
Record in observation	14,205	2,679	0	0
Not missing record date	14,205	2,679	0	0
Merge overlapping records	13,908	2,679	297	0
Restricted to first entry	2,679	2,679	11,229	0
Age requirement: 18 to 150	309	309	2,370	2,370
Sex requirement: Female	176	176	133	133
Prior observation requirement: 30 days	176	176	0	0
Future observation requirement: 0 days	176	176	0	0
In table visit_occurrence between 0 & 0 days relative to cohort_start_date between 1 and Inf	1	1	175	175



Your turn!



Exercise 2 - Requirement and filtering

Create a new cohort named “study_cohort” by applying the following criteria to the base conditions cohort:

- For all cohorts, require that records start between January 1, 1990, and December 31, 2011.
- For the Rheumatoid Arthritis cohort, exclude patients with a history of osteoarthritis prior to or on the day of diagnosis.

```
osteoarthritisCodes <- list("osteoarthritis" = c(80180L)) |> newCodelist()
```

▶  Click to see solution

Move to the next slide to see the cohort attrition



Exercise 2 - Requirement and filtering

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
Delphi-2M; pneumonia				
Initial qualifying events	783	783	0	0
Record in observation	783	783	0	0
Not missing record date	783	783	0	0
Merge overlapping records	783	783	0	0
cohort_start_date after 1990-01-01	578	578	205	205
cohort_start_date before 2011-12-31	578	578	0	0
Delphi-2M; ra				
Initial qualifying events	676	676	0	0
Record in observation	676	676	0	0
Not missing record date	676	676	0	0
Merge overlapping records	676	676	0	0
cohort_start_date after 1990-01-01	487	487	189	189
cohort_start_date before 2011-12-31	487	487	0	0
Not in concept osteoarthritis between -Inf & 0 days relative to cohort_start_date	487	487	0	0

Update cohort entry and exit

Functions to update cohort start and end dates

- Cohort exit
 - `exitAtObservationEnd()`
 - `exitAtDeath()`
 - `exitAtFirstDate()`
 - `exitAtLastDate()`
- Cohort entry
 - `entryAtFirstDate()`
 - `entryAtLastDate()`
- Trim start and end dates
 - `trimDemographics()`
 - `trimToDateRange()`
 - `trimDuration()`
- Pad start and end dates
 - `padCohortDate()`
 - `padCohortEnd()`
 - `padCohortStart()`

Update cohort entry and exit - Example

- We can trim start and end dates to match demographic requirements.
- For instance, cohort dates can be trimmed so the subject contributes time while:
 - Aged 20 to 40 years old
 - Prior observation of at least 365 days

```
cdm$medications_trimmed <- cdm$medications |>
  trimDemographics(
    ageRange = list(c(20, 40)),
    minPriorObservation = 365,
    name = "medications_trimmed"
  )
```

Update cohort entry and exit - Example

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
acetaminophen				
Initial qualifying events	14,205	2,679	0	0
Record in observation	14,205	2,679	0	0
Not missing record date	14,205	2,679	0	0
Merge overlapping records	13,908	2,679	297	0
Non-missing year of birth	13,908	2,679	0	0
Age requirement: 20 to 40	4,049	2,124	9,859	555
Prior observation requirement: 365 days	4,049	2,124	0	0



Your turn!



Exercise 3 - Update cohort entry and exit

- 1) Since Rheumatoid Arthritis is a chronic condition, define its cohort exit so patients remain in the cohort from first diagnostic date to the end of the patient's observation.
- 2) For Pneumonia, assume the infection lasts for two months from the diagnosis day, thereby update the cohort end date to be exactly 60 days after the cohort start.

👉 Use the *cohortId* argument to choose which cohort to modify.

▶ 💡 Click to see solution

Move to the next slide to see the attrition.



Exercise 3 - Update cohort entry and exit

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
Delphi-2M; pneumonia				
Initial qualifying events	783	783	0	0
Record in observation	783	783	0	0
Not missing record date	783	783	0	0
Merge overlapping records	783	783	0	0
cohort_start_date after 1990-01-01	578	578	205	205
cohort_start_date before 2011-12-31	578	578	0	0
pad `cohort_end_date` 60 days	578	578	0	0
Delphi-2M; ra				
Initial qualifying events	676	676	0	0
Record in observation	676	676	0	0
Not missing record date	676	676	0	0
Merge overlapping records	676	676	0	0
cohort_start_date after 1990-01-01	487	487	189	189
cohort_start_date before 2011-12-31	487	487	0	0
Not in concept osteoarthritis between -Inf & 0 days relative to cohort_start_date	487	487	0	0
Exit at observation period end date	487	487	0	0

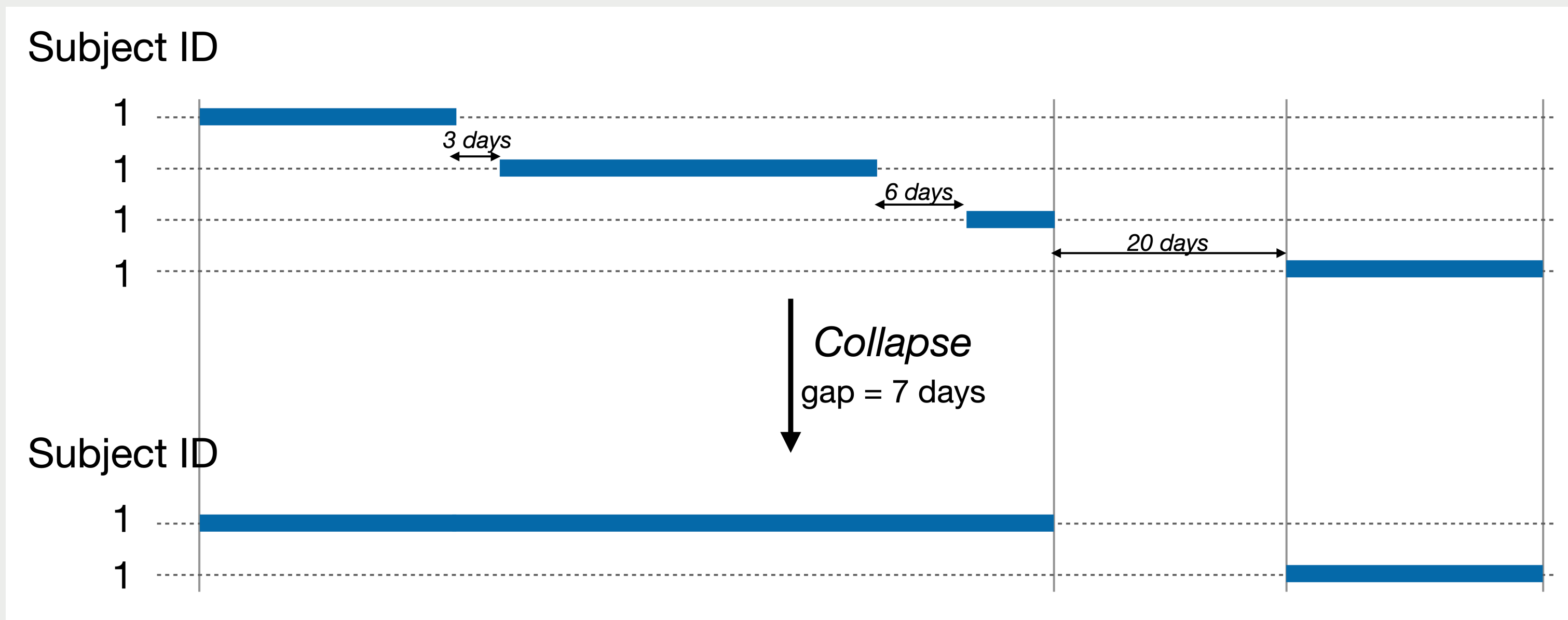
Transformation and Combination

Functions for Cohort Transformation and Combination

- Split cohorts
 - `yearCohorts()`
 - `stratifyCohorts()`
 - `timeWindowCohorts()`
- Combine cohorts
 - `unionCohorts()`
 - `intersectCohorts()`
- Filter cohorts
 - `subsetCohorts()`
 - `sampleCohorts()`
- Match cohorts
 - `matchCohorts()`
- Concatenate entries
 - `collapseCohorts()`
- Copy and rename cohorts
 - `renameCohort()`
 - `copyCohorts()`

Cohort combinations - Example

1. Collapse drug exposures that are within a gap of 7 days.



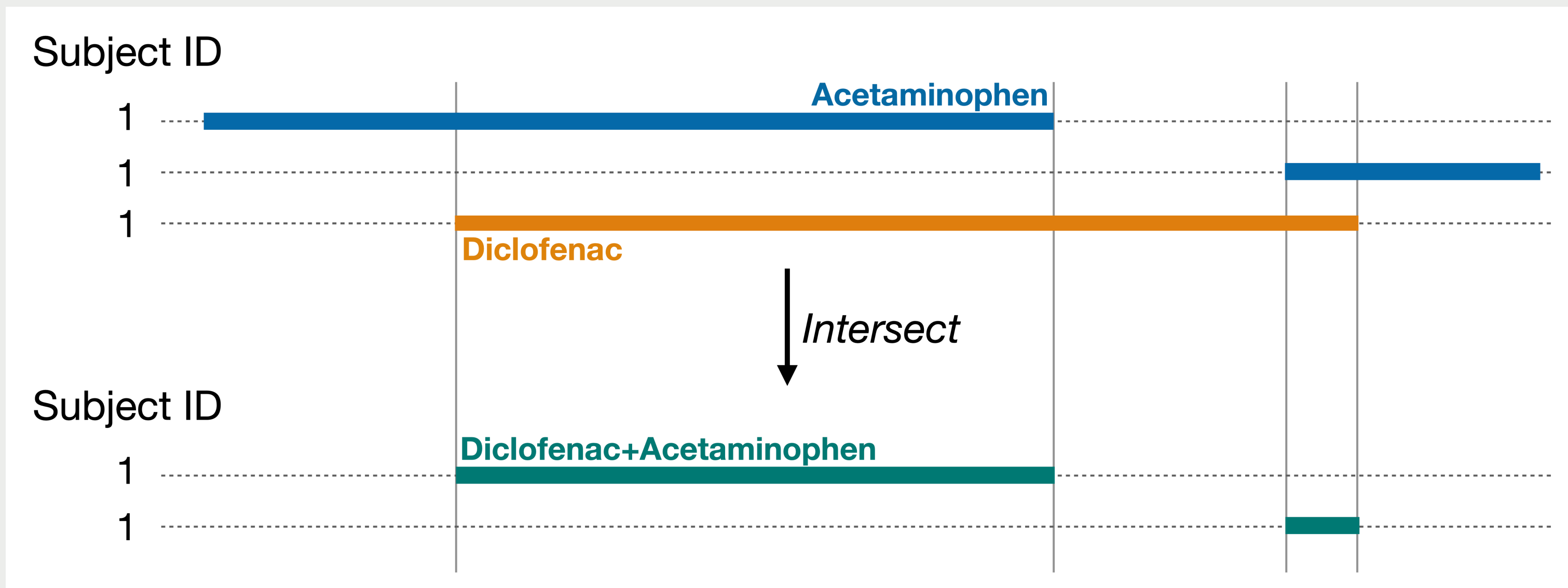
Cohort combinations - Example

```
cdm$medications_collapsed <- cdm$medications |>
  collapseCohorts(gap = 7, name = "medications_collapsed")
```

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
acetaminophen				
Initial qualifying events	14,205	2,679	0	0
Record in observation	14,205	2,679	0	0
Not missing record date	14,205	2,679	0	0
Merge overlapping records	13,908	2,679	297	0
Collapse cohort with a gap of 7 days.	13,899	2,679	9	0

Cohort combinations - Example

2. Create a new cohort that contains people who had an exposure to both **diclofenac** and **acetaminophen** at the same time.



Cohort combinations - Example

```
cdm$intersection <- cdm$medications_collapsed |>
  intersectCohorts(name = "intersection")

settings(x = cdm$intersection)
```

A tibble: 1 × 5

	cohort_definition_id	cohort_name	gap	acetaminophen	diclofenac
	<int>	<chr>	<dbl>	<dbl>	<dbl>
1	1	acetaminophen_diclofenac	0	1	1



Your turn!

Exercise 4 - Transformation and Combination

Create a third cohort containing patients with Pneumonia who have Rheumatoid Arthritis at the same time. The final cohort table should keep all original cohorts plus the new intersection cohort.

Finally, rename cohorts so they are called “pneumonia”, “rheumatoid_arthritis”, and “pneumonia_and_rheumatoid_arthritis”.

▶  Click to see solution

Move to the next slide to see the attrition.



Exercise 4 - Transformation and Combination

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
Delphi-2M; pneumonia				
Initial qualifying events	783	783	0	0
Record in observation	783	783	0	0
Not missing record date	783	783	0	0
Merge overlapping records	783	783	0	0
cohort_start_date after 1990-01-01	578	578	205	205
cohort_start_date before 2011-12-31	578	578	0	0
pad `cohort_end_date` 60 days	578	578	0	0
Delphi-2M; rheumatoid_arthritis				
Initial qualifying events	676	676	0	0
Record in observation	676	676	0	0
Not missing record date	676	676	0	0
Merge overlapping records	676	676	0	0
cohort_start_date after 1990-01-01	487	487	189	189
cohort_start_date before 2011-12-31	487	487	0	0
Not in concept osteoarthritis between -Inf & 0 days relative to cohort_start_date	487	487	0	0
Exit at observation period end date	487	487	0	0
Delphi-2M; pneumonia_and_rheumatoid_arthritis				
Initial qualifying events	0	0	0	0

CohortConstructor

👉 Packages website

👉 CRAN link

👉 Manual



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